

EE186 Homework 2

Due: October 14, 2026 (10:00 PM)

1. Suppose $r0 = 0x20008000$, and the memory layout is as follows: If the system is little

Address	Data
0x20008007	0x79
0x20008006	0xCD
0x20008005	0xA3
0x20008004	0xFD
0x20008003	0x00
0x20008002	0xEB
0x20008001	0x2C
0x20008000	0x1A

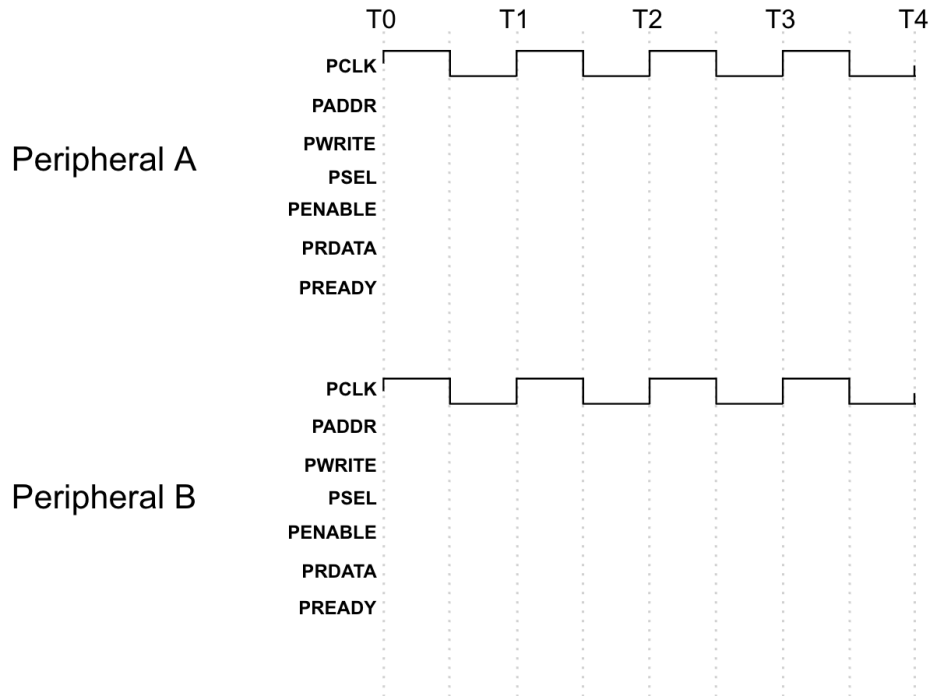
endian, what are the values of $r1$ and $r0$ in each case if the following instructions are executed separately?

- (a) LDR $r1, [r0, \#4]$
 - (b) LDR $r1, [r0], \#4$
 - (c) LDR $r1, [r0, \#4] !$
2. Consider an APB bus system with one Initiator and two Target devices:
- Peripheral A: mapped to addresses $0x00002000 - 0x0000200F$
 - Peripheral B: mapped to addresses $0x00002010 - 0x0000201F$

Assume $r3 = 0x00002000$. The Cortex-M4 processor executes the following instruction:

LDR $r2, [r3, \#0x14]$

- (a) Which peripheral is selected by this instruction?
- (b) Is the processor performing a read or write transfer on the APB bus?
- (c) Sketch the APB timing diagram for this transfer. Assume the bus has no wait states (PREADY of the selected peripheral is guaranteed to be high when PEN-ABLE is set high).



3. Explain why MMIO registers should generally be declared as `volatile` pointers in C.
4. Suppose the SysTick interrupt handler, shown below, is at address 0x0800001C. The SysTick interrupt occurs when SP = 0x20000600, PSR = 0x00000020, PC = 0x08000020, LR = 0x20008020, and register Ri = i, where i = 0, 1, 2, ... , 12.

```

SysTick_Handler PROC
    PUSH {r4, r5, r6}
    ADD r0, r0, #1
    ADD r1, r1, #1
    ADD r2, r2, #1
    ADD r3, r3, #1
    ADD r4, r4, #1
    ADD r5, r5, #1
    ADD r6, r6, #1
    ADD r7, r7, #1
    ADD r8, r8, #1
    ADD r9, r9, #1
    ADD r10, r10, #1
    ADD r11, r11, #1
    ADD r12, r12, #1
    POP {r4, r5, r6}
    BX LR
    ENDP

```

- (a) Show the stack contents and the value of PC and SP immediately after entering the SysTick interrupt service routine, before PUSH instruction is executed.
 - (b) What are the values of these registers (R0-R12, LR, SP, PSR, and PC) immediately after the interrupt exits?
5. A Cortex-M4 MCU is running with three peripherals generating interrupts:
- Timer1 $\rightarrow$ priority 2
 - USART $\rightarrow$ priority 4
 - GPIO $\rightarrow$ priority 1

Assume that the processor is currently executing the main program. Timer1 asserts its interrupt. While its ISR is executing, GPIO and USART interrupts are asserted almost simultaneously. Describe the exact sequence of ISR executions, considering preemption, tail-chaining, and the inverse priority scheme of the NVIC.